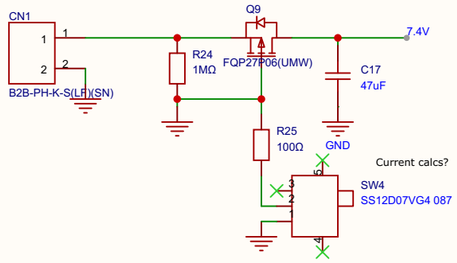
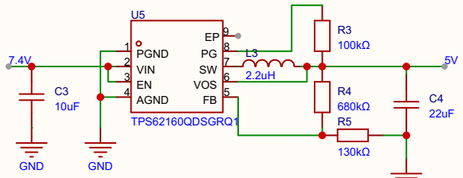
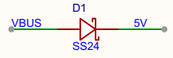
Power input and regulation

(This is the battery, idk which connector for our 2S LiPos)

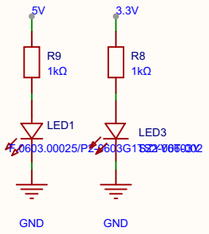


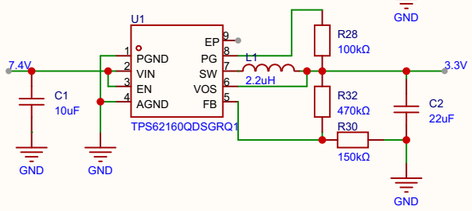
R3&R4:680k and 130k shown on P19 of datasheet for 5V output





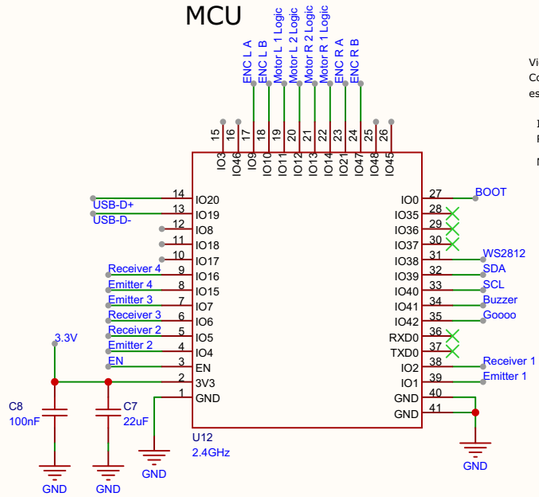
5V and 3.3V indicator LEDs





Schotty Diode to protect USB/Computer. Voltage drop is max 0.55V at 2A, IR might be a tiny sad when debugging.

MCU



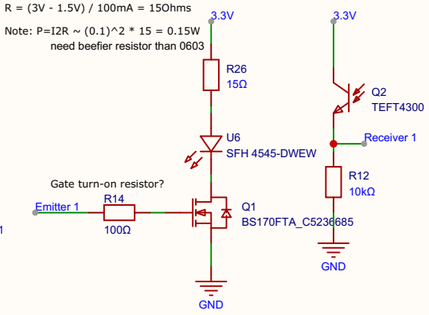
Do not use GPIO35-37
GPIO0 and GPIO46 are boot strapping pins
GPIO45 is a strapping pin for VDD_SPI voltage
GPIO46 straps pin for ROM message printing
GPIO3 straps the JTAG signal source.
GPIO39 goes HIGH after reset, as opposed to the majority of GPIOs going LOW
GPIO42, GPIO47, GPIO21 transition low on reset slower than other GPIOs
Source: https://makerspet.com/blog/esp32-s3-gpio-limitations/

IR Module

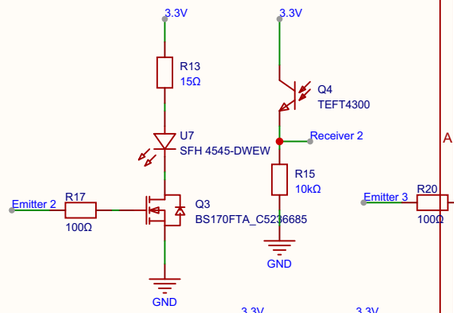
View the phototransistor as a current source (varies with light received)
Could be 10k or 4.7k, depends on light intensity (V=IR, chop to 0.3V~3.1V)
esp32s analog pins are notoriously nonlinear according to AI

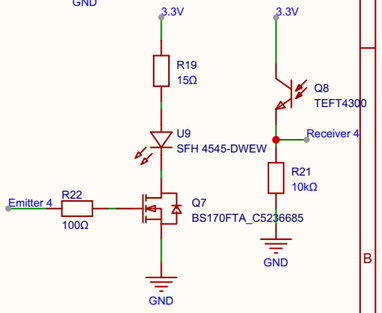
I = 100mA V_LED ~ 1.5V
R = (3V - 1.5V) / 100mA = 15Ohms

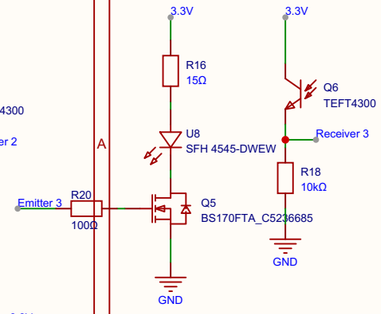
Note: P=I2R ~ (0.1)^2 * 15 = 0.15W
need beefier resistor than 0603







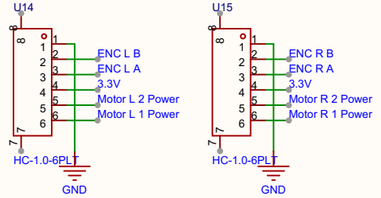




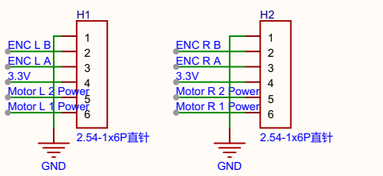
Motor connectors

(Looks like an absolute horror to solder)

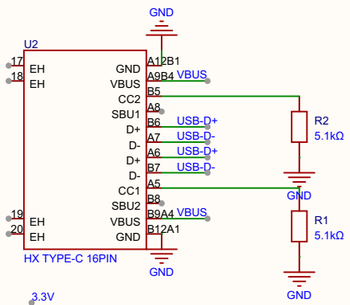
Encoder VCC can be from 2.7-18V so change to 5V if easier routing

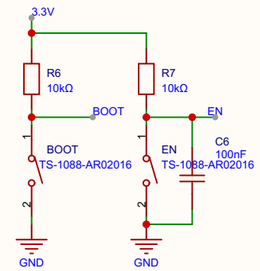


In case we use the polulu boards again/want to swap pin order



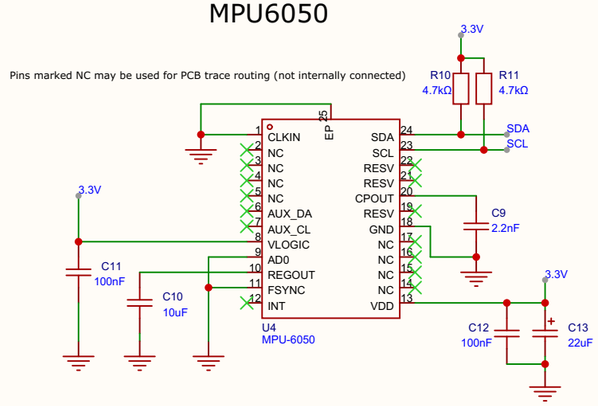
USB & Programming





MPU6050

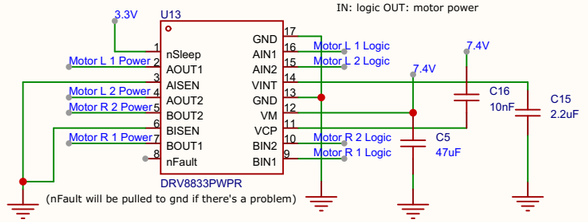
Pins marked NC may be used for PCB trace routing (not internally connected)



Motor Driver

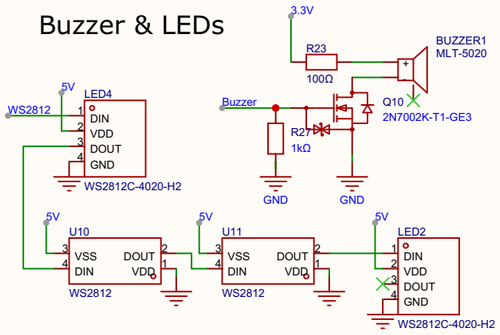
XISEN pins sense voltage drop and will cut off motor signal when I (V=IR) exceeds 0.2V
Polulu max stall current is 1.5A **NVM I'm not adding current sensing resistors**
R = 0.2V / 1.5A ~ 0.133Ohms, use 0.12Ohms which cuts current at 1.66A
P = I2R = 1.66*1.66*0.12 = 0.33W, calls for a beefier resistor

IN: logic OUT: motor power



(nFault will be pulled to gnd if there's a problem)

Buzzer & LEDs



原理图	Firebending	创建日期	2026-07-29
		更新日期	2026-10-25
板子	Zuko V1	图页	P1
绘制		Micromouse	
审阅			
		版本	V1.0
		尺寸	A4
嘉立创EDA		页 1 共 1	嘉立创EDA